HAL-MLE LOG-SPLINES DENSITY ESTIMATION (PART I: UNIVARIATE)

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We study nonparametric maximum likelihood estimation of probability densities under a total variation (TV)–type penalty, namely the sectional variation norm (SVN, also known as Hardy–Krause variation). TV regularization has a long history in regression and density estimation. Here, we revisit this task using the Highly Adaptive Lasso (HAL) framework. We formulate a HAL-based maximum likelihood estimator (HAL-MLE) using the log-spline link function from Kooperberg and Stone (1992), and show that in the univariate setting the bounded SVN assumption underlying HAL coincides with the classical bounded TV assumption. This equivalence directly connects HAL-MLE to existing TV-penalized approaches such as local adaptive splines (Mammen and Van De Geer, 1997). We establish three theoretical results: (i) the univariate HAL-MLE is asymptotically linear, (ii) it admits pointwise asymptotic normality, and (iii) it achieves uniform convergence at rate $n^{-(k+1)/(2k+3)}$ up to logarithmic factors for the smoothness order $k \geq 1$. These results extend existing results from van der Laan and Bibaut (2017), which previously guaranteed only uniform consistency without rates when $k = 0$. We provide a unified framework for TV-penalized density estimation and connect HAL-MLE to existing univariate TV-penalized methods despite its general multivariate formulation.

1. Introduction. We consider nonparametric density estimation on a compact support. Given a random variable $X \sim P_0$, where P_0 is absolutely continuous, and i.i.d. samples $x_{1:n}$ from P_0, our goal is to estimate the true density function p_0 of the underlying distribution P_0. This paper aims to provide a thorough theoretical analysis of univariate density estimation with a variational penalty and its application. The statistical model for P_0 will be nonparametric up till assuming that p_0 has support on a known interval $[a, b]$ and is a cadlag function with a bounded variation norm, explained in detail below. Our framework is naturally extended to the multivariate case.

Nonparametric density estimation methods typically include kernel-based approaches, splines, and wavelet techniques. Kernel density estimation (KDE), despite its simplicity, suffers from several drawbacks: it poorly captures densities with rapidly varying regions and faces severe challenges due to the curse of dimensionality in the multivariate case.

The first drawback of KDE arises from the fact that it is a linear smoother whose fitted values depend linearly on the observed responses. TV penalties have been introduced into spline regression to resolve the same problem with kernel regression, smoothing splines, etc. Mammen and Van De Geer (1997) developed local adaptive splines (LAS) demonstrating L^2 convergence, while Tibshirani (2014) formulated restricted LAS and trend filtering (TF) as general Lasso regressions. Later, the TV penalty was also introduced into the density estimation problem (Bak et al., 2021; Sadhanala et al., 2024). The TV-penalized logspline density estimation (PLSDE) proposed in Bak et al. (2021) is more relevant to our problem,

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and it intends to tackle the oscillation issue of logspines method (Kooperberg and Stone, 1991, 1992). Kooperberg and Stone (1992) employs an exponential link transformation ensuring positivity:

$$p_f(x) = \frac{\exp(f(x))}{\int_a^b \exp(f(u))d\mu(u)}, \tag{1.1}$$

with splines $f(x) = \sum_j \beta_j \phi_j(x)$, typically cubic B-splines. Parameters are estimated via maximum likelihood, often guided by criteria such as AIC or BIC. A theoretical analysis of the logspline model is provided in Stone (1990). Bak et al. (2021) employed TV penalty and BIC criteria to provide univariate KL-divergence convergence analysis and its generalization to bivariate case. However, their method, Penalized Log-spline Density Estimation (PLSDE), encounters difficulties generalizing to multivariate settings without assuming higher-order continuity. This is a manifestation of the curse of dimensionality, and it falls into the same problem for the multivariate spline when the TV penalty is not introduced. Another problem is that PLSDE is limited to the uniform knots, which is not preferred in the multivariate case. An argument of the knot placement problem is provided in Section 2.

In contrast, the Highly Adaptive Lasso (HAL) proposed by van der Laan (2017) and practically introduced in Benkeser and Van Der Laan (2016) estimates multivariate càdlàg functions defined on $[0,1]^d$ using a sectional variation norm, yielding uniform consistency and pointwise asymptotic normality without dimension-enforced smoothness assumptions. The same function class has received broad attention in other settings to avoid the curse of dimensionality asymptotically, including nonparametric isotonic/variation-constrained regression (Fang, Guntuboyina and Sen, 2021) and MARS-type regression (Ki, Fang and Guntuboyina, 2024); in this paper, we focus on nonparametric density estimation. The density estimator based on HAL is referred to as HAL-MLE log-splines method. In this paper, we restrict our focus to univariate HAL-MLE in order to compare the performance with the classical log-spines methods and demonstrate theoretical connections to LAS and TF approaches, even though the HAL theory applies to the multivariate case. Other than density estimation, HAL-MLE also provide the asymptotic efficiency guarantee for general pathwise-differentiable statistical estimand, i.e. moments, survival probability, or percentiles, by a simple plug-in or a single-step TMLE procedure as developed in van der Laan (2017); van der Laan, Benkeser and Cai (2023).

The remainder of this paper is organized as follows. Section 2 introduces the HAL assumption and establishes its connection to the classical bounded total variation (BTV) assumption underlying local adaptive splines. Section 3 presents the construction of HAL-MLE with the log-spline link function (Leonard, 1978; Silverman, 1982; Kooperberg and Stone, 1992; Rytgaard, Eriksson and van der Laan, 2023). Section 4 presents the theoretical results on its univariate L^2 convergence, asymptotic linearity, pointwise asymptotic normality, and uniform convergence. We also propose a variance estimator for the density based on the delta method. Section 5 considers the plug-in HAL-MLE and HAL-TMLE of pathwise differentiable statistical estimands, shown to achieve asymptotic efficiency with influence-curve-based variance estimation. Section 6 reports simulation results, empirically verifying the theoretical guarantees mentioned in the previous sections, and comparing the finite sample performance of HAL-MLE with TF (applied to density estimation), log-splines, and KDE. In Section 7, we present a case study with the galaxy velocity data to visualize the convergence, confidence intervals, and targetings. Additional computational details and optimization algorithm performance analyses are provided in Section S.10 of the Supplementary Material (Hou et al., 2026). We provide a Python package for HAL-MLE density estimation, available at https://github.com/zhengpu-berkeley/HALDensity, and experiment details at https://github.com/yilongHou/link_HAL_MLE.

2. The Highly Adaptive Lasso (HAL) Assumptions. The HAL assumptions refer to *càdlàg* functions with bounded sectional variational norm (BSVN), and the modeling of HAL is a natural extension from the exact representation of this function class. In the univariate case, a *càdlàg* function could be considered as a function that can be approximated by a linear combination of simple indicator functions that jumps from 0 to 1 at a knot-point. The function class includes the linear combination of cumulative distribution functions. Gill, van der Laan and Wellner (2001) showed that the concept of SVN extends the Donsker property of univariate *càdlàg* functions with bounded total variation to multivariate cases. Some follow-up works of this results appear in van der Laan (2017); Rytgaard, Eriksson and van der Laan (2023). A detailed discussion on *càdlàg* functions with BSVN and their measurability is provided in Munch et al. (2024, Section 2). A similar result also appears in Radulović, Wegkamp and Zhao (2017). And in some literature like Radulović, Wegkamp and Zhao (2017); Ki, Fang and Guntuboyina (2024), the SVN is referred to as the Hardy-Krause variation. We here introduce these concepts and their relationship to HAL.

2.1. *Basic Concepts.* We first introduce the function class of *càdlàg* functions with BSVN (upper bounded by some $U < \infty$) without assuming any order of differentiability, denoted as $D_U^{(0)}([0,1]^d)$.

DEFINITION 2.1 (*Càdlàg* Functions with Bounded Sectional Variational Norm (BSVN)). The function $f \in D_U^{(0)}([0,1]^d)$ is the function that can be represented as follows:

$$f(x) = f(0) + \sum_{s \subset \{1,\ldots,d\}} \int_{(0(s),\, x(s)]} f\big(du_s, 0(-s)\big),$$

where s is any nonempty subset of $\{1,\ldots,d\}$, and $-s \equiv \{1,\ldots,d\} \setminus s$. Here, $x(s)$ denotes the components of x with indices in s, while $0(s)$ is the zero vector of the same length as $x(s)$, so that $(0(s), x(s)]$ is a left-open right-closed cube within the $|s|$-dimensional unit cube. Accordingly, the SVN is defined by

$$\|f\|_v^* = \big|f(0)\big| + \sum_{s \subset \{1,\ldots,d\}} \int_{(0(s),\, x(s)]} \Big| f\big(du(s), 0(-s)\big)\Big|.$$

This is a stronger assumption than the bounded total variation (BTV) assumption, since it requires bounded variation not only on the full coordinate set $s = \{1,\ldots,d\}$ but on all lower-dimensional sections $s \subset \{1,\ldots,d\}$.

In the univariate case $D_U^{(0)}([0,1])$, these definitions simplify to

$$f(x) = f(0) + \int_{(0,x]} f(du), \quad \|f\|_v^* = \big|f(0)\big| + \int_{(0,x]} \Big|f(du)\Big|. \tag{2.1}$$

When f is also k-th order differentiable, we define the k-th order sectional variational norm, $\|f\|_{v,k}^*$, and the function subclass of $D_U^{(0)}([0,1])$ with bounded $\|f\|_{v,k}^*$, $D_U^{(k)}([0,1])$, as follows:

DEFINITION 2.2 (The Function with k-th order BSVN). For the smoothness order $k \geq 1$ and a large enough constant $U > 0$, we say $f \in D_U^{(k)}([0,1])$ if the following hold:

1. $f \in D_U^{(0)}([0,1])$.

2. For each $1 \le i \le k$, the i-th Lebesgue–Radon–Nikodym derivative of f exists
$$f^{(i)}(u) \;=\; \frac{\mathrm{d}}{\mathrm{d}u} f^{(i-1)}(u)$$
and $f^{(i)} \in D_U^{(0)}([0,1])$.
3. The k-th order sectional variation norm of f, in the univariate case, simplified as
$$\|f\|_{v,k}^* \;= \sum_{i=0}^{k} |f^{(i)}(0)| + \mathrm{TV}\big(f^{(k)}\big),$$
satisfies
$$\|f\|_{v,k}^* \;\le\; U, \text{for some } U < \infty$$

A quick note on the notation: We use $\mathrm{TV}(\cdot)$ to denote the total variation functional as defined in Section 9.1 of Tibshirani et al. (2022). When the function f is a càdlàg function, $\mathrm{TV}(f) = \int_0^1 |df(u)|$. This is also referred to as the total variation norm (a seminorm), denoted as $|f|_v$. We then write $(x-u)_+^k = (x-u)^k I\{u \le x\}$ where I is the indicator, and this truncated k-th order spline with knot point $u \in [0,1]$ is k-th order weakly differentiable. Notice that the following representation theorem and its derivation are specialized to the univariate case, and we include it here because it motivates and unifies the parametrization of such a function class. Here we demonstrate the exact representation for $f \in D_U^{(k)}([0,1])$ with the concepts above.

PROPOSITION 2.3 (Univariate Exact Representation). *Any $f \in D_U^{(k)}([0,1])$ could be represented as follows:*
$$f(x) = \sum_{i=0}^{k} \frac{1}{i!} f^{(i)}(0)\, x^i + \int_0^1 \frac{1}{k!} (x-u_k)_+^k \, d\, f^{(k)}(u_k).$$

The multivariate or higher-order spline exact representation is included in van der Laan (2023, Section 3).

2.2. *Relationship with TV.* BTV is a classic assumption of spline methods for the regression problem. The implementation is to derive TV as a function of the coefficients of the bases. Thus, different basis systems have different representations of the TV. The HAL assumptions seem to be stricter by Definition 2.1, but there is actually no loss comparing to the BTV assumption in the univariate case. LAS uses the same truncated power basis as in univariate HAL, and therefore we argue the equivalency of the two assumptions through analyzing the estimation of LAS.

2.2.1. *A Review of Locally Adaptive Regression Splines (LAS).* Suppose f is a k-th order differentiable function, and $f^{(k)}$ is the k-th order derivative of f. By convention $f^{(0)} \equiv f$, or WLOG we can adopt the term that $f^{(k)}$ is the k-th weak derivative of f from Tibshirani (2014). Mammen and Van De Geer (1997) proposed LAS, which achieved optimal L^2 convergence rates in $\mathcal{F}_k$, where
$$\mathcal{F}_k \;=\; \big\{\, f : [0,1] \to \mathbb{R} \;\big|\; f \text{ is } k\text{-th order differentiable and } \mathrm{TV}\big(f^{(k)}\big) < \infty \big\}.$$

The optimization over $\mathcal{F}_k$ is referred to as the unrestricted LAS method as follows:

$$\hat{f} \in \arg\min_{f \in \mathcal{F}_k} \frac{1}{2} \sum_{i=1}^{n} \big(y_i - f(x_i)\big)^2 + \lambda \, \mathrm{TV}\big(f^{(k)}\big), \tag{2.2}$$

Due to the difficulty of implementation, especially when $k \geq 2$, they proposed the restricted LAS method as an asymptotic solution, as follows:

$$\mathcal{F}_{n,k} = \Big\{ f : [0,1] \to \mathbb{R} \,\Big|\, f \text{ is a linear combination of } k\text{-th order splines with knots in } x_{1:n} \text{ and } \mathrm{TV}\big(f^{(k)}\big) < \infty \Big\},$$

and the optimization problem becomes

$$\hat{f} \in \arg\min_{f \in \mathcal{F}_{n,k}} \frac{1}{2} \sum_{i=1}^{n} \big(y_i - f(x_i)\big)^2 + \lambda \, \mathrm{TV}\big(f^{(k)}\big). \tag{2.3}$$

Mammen and Van De Geer (1997) showed the asymptotic convergence of the solutions of the two problems when the distances among $x_{1:n}$ are close enough. In Mammen and Van De Geer (1997, Theorem 9), they showed that, for the truth $g_{0,n}$ in a generic function class $\mathcal{G}$ with BTV, if one can choose a good enough linear subspace $\mathcal{G}_{n,k}$ with an oracle $g_{1,n}$, then the solution $\hat{g}$ on $\mathcal{G}_{n,k}$ preserves the optimal L^2 convergence on $\mathcal{G}$. Thus, the asymptotic conclusion holds by choosing $\mathcal{G}$ with BTV to be $\mathcal{F}_k$, and $\mathcal{G}_{n,k}$ to be $\mathcal{F}_{n,k}$.

2.2.2. *The Missing* Càdlàg *Perspective.* As discussed, the restricted LAS uses a finite linear combination of truncated power basis as the estimation, and thus the estimation is always a *càdlàg* function. Its generalization to non-*càdlàg* functions is achieved through the equivalency of a class of functions with the same TV. In order words, one can make a function into a *càdlàg* function by shifting finite points while maintaining the same TV, and the LAS provides the same estimation. The L^2 convergence still holds for this finite pointwise difference. For example, $I(x < 1)$ is left continuous, moving the point at $x = 1$ up to 1 provides $I(x \leq 1)$ which is *càdlàg*, and they are indistinguishable under the same TV penalty. That is to say, among this set of indistinguishable functions, there must be a *càdlàg* function. And the convergence to non-*càdlàg* functions achieved by the LAS ignores pointwise behavior.

We claim that, by the representation in Proposition 2.3, $D_U^{(k)}([0,1])$ is the closure of the infinite linear combination of k-th order truncated power splines. Thus, we conclude that the solution based on the finite-dimensional working model $\mathcal{F}_{n,k}$ is a finite-dimensional approximation of f in $D_U^{(k)}([0,1])$, where f is the indistinguishable *càdlàg* function of the truth f_0. Noticing that the pointwise behavior does not hurt L^2 convergence, but we need the rigorous *càdlàg* assumption for proving uniform convergence in Section 4.

2.2.3. *Conclusion.* In the univariate case, when $k = 0$, the SVN only requires the existence of $|f(0)|$ and BTV. And when $k \geq 1$, the *càdlàg* assumption and the existence of $|f^{(i)}(0)|$ for all $1 \leq i \leq k$ is implied by differentiability, and so BSVN and BTV are equivalent assumptions.

REMARK 2.4. We could argue that we only need the Lebesgue–Radon–Nikodym derivative, and relax our assumption by a null set with Lebesgue measure 0. But they could be augmented in a similar manner as well. This is the reason Tibshirani (2014) introduced weak differentiability for the function class and the working model in Section 3 therein.

3. HAL Density Estimation. With the HAL assumptions and the same link function in Kooperberg and Stone (1992), we here form the density estimation technique, HAL-MLE. The intuition of this choice of link function is to assume that the log-likelihood of the density follows a càdlàg structure with bounded sectional variation, making it well-suited for modeling via HAL. This link function is also referred to as the histo-spline method (Leonard, 1978; Silverman, 1982). Specifically, when using zero-order HAL basis functions, the function f can be expressed as a histogram. One major benefit of this link function is that the resulting density belongs to the exponential family, ensuring the convexity of the MLE optimization with L_1 norm regularization.

3.1. *HAL Construction.* Proposition 2.3 shows that we can represent any function $f \in D_U^{(k)}([0,1])$ by an infinite linear combination of the k-th order splines (bases). We then could construct our estimation with a finite-dimensional working model by approximating the integral with a sum. This explains the bias of HAL model by not including all the bases. We initialize the pre-selection working model by partitioning the integral into n parts with knot points $x_1, \cdots, x_n$. Let $f_0 \in D_U^{(k)}([0,1])$ be the truth, and f_n be the estimation based on the n-knot points. Note that the dimension of the working model here refers to the parameters needed for modeling rather than the structure of observations.

$$f_n(x) = \sum_{i=0}^{k} \frac{f^{(i)}(0)}{i!}\, x^i + \sum_{j=1}^{n} \frac{\Delta\, f^{(k)}(x_j)}{k!}\, (x - x_j)_+^k\, .$$

And we could summarize the basis system as follows:

$$\phi_{i,0}(x) = x^i, \quad i = 0, \dots, k,$$

$$\phi_{k,x_j}(x) = (x - x_j)_+^k, \quad j = 1, \dots, n.$$

This basis naturally divides into:

- The **parametric (global) part**: the global polynomial trend (x^i-type terms),
- The **nonparametric (local) part**: the truncated power functions that allow flexible deviations at specified knots $x_{1:n}$.

REMARK 3.1. Such a basis system is also called the truncated power basis (Schumaker, 2007, Theorem 8.51), and each basis can be represented by a tuple of basis order and knot points, i.e., for the nonparametric part of the basis system, we index them each by a tuple (k, u) for k-th order splines and any knot point $u \in (0, 1]$. For the parametric part of the basis system, we index them each by a tuple $(i, 0)$, for $i \leq k$ and starting point 0.

We denote this working model index set as $\mathcal{R}^k(n)$ with cardinality $n + k + 1$. This is referred to as the initial working model, whereas the model after Lasso selection is called the post-selection working model, denoted as $\mathcal{R}_n$ with cardinality J_n. Notice that the initial working model involves $(n + k + 1)$ parameters, and an element in this working model is of the form

$$f_{n,\beta}(x) = \sum_{i=0}^{k} \beta_i\, \phi_{i,0}(x) + \sum_{j=1}^{n} \beta_{k+j}\, \phi_{k,x_j}(x).$$

Hence, its k-th order sectional variation norm is $\left\|f_{n,\beta}\right\|_{v,k}^{*}=\|\beta\|_1$. And for a general loss L, the empirical risk minimizer is

$$\beta_n(M)=\arg\min_{\|\beta\|_1\leq M} P_n\, L\big(f_{n,\beta}\big), \qquad f_n(x)=f_{n,\beta_n(M)}(x). \tag{3.1}$$

M here serves as a user-specified hyperparameter as the L_1 norm constraint of the basis coefficients. In practice, it could be tuned by cross validation and the corresponding choice of M is denoted as M_{cv}, and the process is referred to as CV-HAL-MLE. The undersmoothening of HAL-MLE is achieved by choosing a $M'>M_{cv}$, the post-selection working model enlarges, but still indexed by the initial working model. The undersmoothened HAL-MLE reduces bias by including more bases, but increases variance as a trade-off. See Section S.1 of the Supplementary Material (Hou et al., 2026) for more detailed definitions.

3.2. *HAL-MLE with Link Function.* As the setup in the introduction, let the $X\sim P_0$ be defined on $[0,1]$. We index the density $p_0=\mathrm{d}P_0/\mathrm{d}\mu$ in terms of a p_{f_0} for a function $f_0\in D_U^{(k)}([0,1])$, where μ could be chosen as the Lebesgue measure for simplicity. Thus, the statistical model is defined as $\mathcal{M}=\{p_f : f\in D_U^{(k)}([0,1])\}$. The HAL-MLE of is then denoted as $f_{n,\beta_n(M)}$, and $\beta_n(M)$ is estimated as follows:

$$\beta_n(M)=\arg\min_{\beta:\|\beta\|_1<M}\sum_{i=1}^{n} L(f_{n,\beta})(X_i)=\arg\max_{\beta:\|\beta\|_1\leq M}\sum_{i=1}^{n}\sum_{j=1}^{n+k+1}\beta_j\phi_j(x_i)-n\log C(\beta),$$

where $x_{1:n}\sim P_0, i.i.d.$, $L=-\log p_f$ is the loss function, and the $C(\beta)=\int_0^1\exp(f_\beta(x))\mathrm{d}x$ is the normalizing constant.

REMARK 3.2. This convex optimization problem can be solved by generic optimization solvers like CVXPY. However, the major drawback is that the knot points selection is not as effective as the sample size increases. We propose a list of standard algorithms, including fast iterative shrinkage-thresholding algorithm (FISTA), projected gradient descent, and proximal Newton method and quasi-Newton method (L-BFGS), etc. Generally speaking, comparing to the CVXPY solvers, the optimization algorithms we proposed succeed in selecting knot points and converging to the global optimum. This aligns with the HAL-MLE theory of the size of the post-selection working model, as discussed in Sections S.2 and S.5 of the Supplementary Material (Hou et al., 2026), that we only need $O^{+}(n^{\frac{1}{2k+3}})$ knots to achieve the desired performance. We provide additional optimization details in Section S.10 of the Supplementary Material (Hou et al., 2026).

3.2.1. *Comparison with NPMLE.* Our set $\mathcal{R}^k(n)$ of k-th order spline basis functions is determined by data and its linear span is flexible enough to provide a strong approximation of any function in our class $D_U^{(k)}([0,1])$. Moreover, for large enough U, the latter class will contain the true f_0 identifying the true density p_{f_0} of X. Therefore, the HAL-MLE over the linear combination of basis in $\mathcal{R}_n$ with some L_1 norm upper bound, denoted as $D_U^{(k)}(\mathcal{R}_n)$ approximates an MLE over the full class $D_U^{(k)}([0,1])$ containing the true f_0. Consequently we can view the HAL-MLE as a Nonparametric Maximum Likelihood Estimation (NPMLE) method for estimating the true density of the data. An important distinction between HAL-MLE and the traditional NPMLE methods, like the empirical distribution and Kaplan–Meier (KM) estimator (Kaplan and Meier, 1958), lies in the functional constraint HAL imposes. HAL-MLE produces density estimates confined to the space $D_U^{(k)}([0,1])$. In contrast, traditional NPMLE imposes no smoothness or variation-norm constraints on the densities. As a consequence, the NPMLE is attained at the (non-density) empirical discrete distribution that puts mass $1/n$ on each observation x_i. This is not even a real density, even when there are reasons to believe that the underlying true density is continuous. Similarly, the NPMLE of the failure time distribution for right-censored data under the assumption of independent censoring is attained at a discrete failure time distribution, while the HAL-MLE analogue would result in a valid density estimator of the true failure time density. NPMLE can still provide efficient plug-in estimators of smooth functionals of the density, such as a survivor function. The HAL-MLE, possibly extended with the HAL-TMLE, also yield efficient plug-in estimators of such smooth features, while it still also estimates the density well.

3.2.2. *Uniform Grid vs Data Adaptive Grid.* Uniform grid is a common choice in the literatures of univariate splines methods. PLSDE (Bak et al., 2021) benefits from this choice for the representation of TV penalty with B-splines. TF (Tibshirani et al., 2022) also benefits from this choice for the simple recursion of the divided difference matrix. And theoretically, both uniform grid and data-adaptive grid could provide a strong approximation of any BTV function.

In HAL-MLE, we prefer data-adaptive initialization rather than uniform knots, even though the matter is nuisance in the univariate case. When generalized to multivariate case, the uniform grid becomes an initialization of n^d knots. However, a d-dimensional function has $2^d - 1$ sections, and HAL initialize $n(2^d - 1)$ knots, n knots per section. We have a much smaller working model to begin with.

4. Theoretical Properties for HAL-MLE. With the HAL assumptions and HAL-MLE setup, we here provide a comprehensive theoretical analysis of the convergence properties of spline density estimation with variational penalty, including L^2-convergence, pointwise asymptotic normality, and uniform consistency.

4.1. *Rate of Convergence of HAL-MLE in Loss-based Dissimilarity.* We first establish the L^2 convergence rate of the HAL-MLE. This result is natural in the context of univariate spline methods penalized by total variation, and similar conclusions have been obtained for related estimators (Mammen and Van De Geer, 1997; Tibshirani, 2014; Bak et al., 2021). Our contribution is to demonstrate the L^2 convergence of the HAL-MLE without any assumptions, using a standard proof strategy based on loss-based dissimilarity. Recall that the loss-based dissimilarity, $d_0(f_n, f_0) \equiv P_0 L(f_n) - P_0 L(f_0)$, coincides with the Kullback–Leibler divergence when $L = -\log p_f$. Let

$$\mathcal{F} := D_U^{(k)}([0,1]), \qquad f_0 := \arg\min_{f \in \mathcal{F}} P_0 L(f),$$

where $D_U^{(k)}([0,1])$ denotes the kth-order bounded variation class on $[0,1]$. For a user-specified L_1-norm constraint M and a finite-dimensional index set $\mathcal{R}_n$, define the empirical estimator

$$f_n := \arg \min_{f \in D_M^{(k)}(\mathcal{R}_n)} P_n L(f).$$

The proof relies on two facts.

1. (Positivity Condition) With the link function, for all $f \in \mathcal{F}$, there exists $\delta > 0$ such that $p_f \geq \delta$ uniformly.
2. (Second-order Loss Behavior Condition) When $L(f) = -\log p_f$ (negative log-likelihood), we have

$$\sup_{f \in \mathcal{F}} \frac{P_0\big\{L(f) - L(f_0)\big\}^2}{d_0(f, f_0)} = O(1) < \infty.$$

REMARK 4.1. For the first fact, note that under the bounded variation assumption, all $f \in \mathcal{F}$ are uniformly bounded. Consequently, the link function is uniformly bounded away from zero and bounded above, so the required positivity condition holds automatically. The second fact follows directly from van der Laan, Dudoit and Keles (2004).

REMARK 4.2. For the ones familiar with the logspline literature, the L^2 and L^∞ convergence of logsplines in Stone (1990) are very different from ours. First, they deal with a parametric MLE on equally spaced knots, whereas we deal with data-adaptive knots.Second, the convergence they showed is between the estimation and the parametric oracle MLE, which is defined as the argmax of the expected value of the parametric log-likelihood.

THEOREM 4.3 (L^2 convergence of HAL-MLE). *If $f_0 \in D_U^{(k)}([0,1])$ and $L(f) = -\log p_f$, then,*

$$d_0(p_{f_n}, p_{f_0}) = O_P\left(n^{-\frac{2k+2}{2k+3}}\right), \quad \textit{and hence} \quad \|p_{f_n} - p_{f_0}\|_{L^2(\mu)} = O_P\left(n^{-\frac{k+1}{2k+3}}\right).$$

This rate agrees with the L^2 convergence rate for order $k = 0$ and dimension $d = 1$ in regression (Mammen and Van De Geer, 1997; Tibshirani, 2014; Bibaut and van der Laan, 2019). Our density estimation setting preserves this rate from the regression setting, which is sharp and without any assumptions.

4.2. *Pointwise Asymptotic Normality of the HAL-MLE.* Recall the link function, $p_\beta = p_{f_\beta} = \frac{\exp(f_\beta)}{C(\beta)}$ where $C(\beta) = \int_0^1 \exp(f_\beta(x))\mathrm{d}x$. The score for the log-likelihood loss is calculated as $S_\beta(\phi_j) = \frac{\mathrm{d}}{\mathrm{d}\beta(j)} \log p_\beta$ at β for the j-th entry $\beta(j)$. Note that this equals

$$S_\beta(\phi_j) = \phi_j(X) - P_\beta \phi_j, \tag{4.1}$$

where $P_\beta \phi_j \equiv \int \phi_j(x) p_\beta(x) \mathrm{d}x = \mathbb{E}_\beta[\phi_j]$. We use the notation $S_\beta(\phi)$ to denote the score vector, and $\phi \equiv (\phi_1, \cdots, \phi_{J_n})^\top$. By the linearity of expectation functional, we note $\phi \to S_\beta(\phi)$ is linear in ϕ.

Recall that $\mathcal{R}_n \subset \mathcal{R}^k(n)$ is a data-adaptive index set of basis functions with cardinality J_n corresponding with the non-zero coefficients in the HAL-MLE fit f_{n,β_n}, using a particular M_n for the L_1-norm constraint. The HAL-MLE implies a J_n-dimensional working model

$$D^{(k)}(\mathcal{R}_n) = \left\{ f_{n,\beta} = \sum_{j \in \mathcal{R}_n} \beta_j \, \phi_j(x) : \beta \right\} \subset D^{(k)}([0,1]).$$

The HAL-MLE $f_n = f_{n,\beta_n}$ also equals a minimizer over this working model $D_{M_n}^{(k)}(\mathcal{R}_n)$ so that

$$f_n = \arg \min_{f \in D_{M_n}^{(k)}(\mathcal{R}_n)} -P_n \log p_f.$$

One nice property of the link function is that the information matrix defined as the negative of the partial derivative of the expectation of the score, $-d/d\beta(i) P_0 S_\beta(\phi_j)$, equals the covariance of the scores $S_\beta(\phi_j) S_\beta(\phi_i)$ under P_β, which can be viewed as an inner product between the two basis functions ϕ_i and ϕ_j. This inner product is an inner product for the linear space $D^{(k)}(\mathcal{R}_n)$ providing an orthonormal basis that will make the information matrix for the reparametrized $D^{(k)}(\mathcal{R}_n)$ a diagonal matrix and while making the scores orthogonal, and equivalently, uncorrelated w.r.t. P_β. This provides us with a powerful ingredient for our formal analysis of the HAL-MLE.

Specifically, for a basis function ϕ_i and its corresponding parameter $\beta(i)$, the i-th element of β,

$$
\begin{aligned}
-\frac{\mathrm{d}}{\mathrm{d}\beta(i)} P_0 S_\beta(\phi_j) &= 0 + \frac{\mathrm{d}}{\mathrm{d}\beta(i)} P_0\{P_\beta[\phi_j]\} = \frac{\mathrm{d}}{\mathrm{d}\beta(i)} P_\beta[\phi_j] \\
&= \int \phi_j(x) \frac{\mathrm{d}}{\mathrm{d}\beta(i)} p_\beta(x)\,\mathrm{d}x = \int \phi_j(x) S_\beta(\phi_i) p_\beta(x)\,\mathrm{d}x \\
&= \int (\phi_j(x) - P_\beta\phi_j) S_\beta(\phi_i) p_\beta(x)\,\mathrm{d}x = P_\beta S_\beta(\phi_j) S_\beta(\phi_i) \\
&\equiv \langle \phi_i, \phi_j \rangle_\beta. \qquad (4.2)
\end{aligned}
$$

W.r.t. the orthonormal basis parametrization of $D^{(k)}(\mathcal{R}_n)$ for which the information matrix is diagonal, we can obtain a straightforward linearization of the HAL-MLE w.r.t. the oracle MLE $f_{\mathcal{R}_n,0} \equiv \arg\min_{f \in D_M^{(k)}(\mathcal{R}_n)} -P_0 \log p_f$ by being able to trivially invert the information matrix, thereby avoiding the delicate understanding of eigenvalues of the original information matrix in terms of the original parametrization of $D^{(k)}(\mathcal{R}_n)$.

Nonetheless, such an analysis runs into the following complication. With this analysis, we can approximate $f_{n,\beta_n} - f_{\mathcal{R}_n,0}$ with an empirical mean of a function of X_i where this function is indexed by $\mathcal{R}_n$, thereby not allowing the application of a central limit theorem. In other words, the data dependence of $\mathcal{R}_n$ makes it hard to push towards a CLT. One could resolve this by using a a priori set of J_n knot-points and select J_n (e.g. uniform knots) with cross-validation. However, the Lasso provides an effective way of selecting the right set of knot-points for fitting the true density so that this would be a practically inferior procedure, and the utility of the Lasso to select the working model is even more important for the extension to multivariate density estimation.

Therefore, for proving the pointwise asymptotic normality of the HAL-MLE, we create an independent version $\mathcal{R}_{n,0}$ of $\mathcal{R}_n$ that is independent of P_n and yields a sup-norm approximation of $D_U^{(k)}([0,1])$ at rate $O(1/J_{n,0}^{k+1})$ up till a $\log n$ factor. Such sets have been shown to exist in van der Laan (2023). We could view $\mathcal{R}_n = \hat{R}(P_n)$ as an algorithm that maps the data P_n into a set of basis functions. Let $P_n^{\#}$ be the empirical probability measure of an independent sample of n i.i.d. observations from P_0. We can then define a particular independent version as $\mathcal{R}_{n,0} = \hat{R}(P_n^{\#})$. We use this independent version to define an independent working model $D^{(k)}(\mathcal{R}_{n,0})$ of dimension $J_{n,0}$, which can be viewed as an approximation of the data dependent working model $D^{(k)}(\mathcal{R}_n)$. This independent working model defines a different oracle MLE

$$
f_{n,0} = f_{\mathcal{R}_{n,0},0} \equiv \arg\min_{f \in D_M^{(k)}(\mathcal{R}_{n,0})} -P_0 \log p_f \;=\; f_{\mathcal{R}_{n,0},\beta_{n,0}}.
$$

To emphasize the two different working models we also use notation $f_{\mathcal{R}_n,\beta}$ and $f_{\mathcal{R}_{n,0},\beta}$ as parametrizations of the two working models $D^{(k)}(\mathcal{R}_n)$ and $D^{(k)}(\mathcal{R}_{n,0})$. Specifically, we also use notation $f_{\mathcal{R}_n,0}$ and $f_{\mathcal{R}_{n,0},0}$ for the two oracle MLEs, where the latter is also denoted with $f_{n,0}$. We now analyze the HAL-MLE w.r.t. the independent oracle MLE $f_{\mathcal{R}_{n,0},0}$ instead of $f_{\mathcal{R}_n,0}$, relying on showing that the HAL-MLE over $D^{(k)}(\mathcal{R}_n)$ indeed can be viewed as an approximate MLE over $D^{(k)}(\mathcal{R}_{n,0})$.

Although f_n is not a full MLE over $D^{(k)}(\mathcal{R}_{n,0})$, it can be shown to act as an approximate MLE solving its score equations at close enough approximation so that f_n not only acts as an MLE over the data dependent working model $D^{(k)}(\mathcal{R}_n)$ but also as an MLE over the independent working model $D^{(k)}(\mathcal{R}_{n,0})$. The approximation error in solving these score equations is a term in our analysis that is formally addressed in Section S.3 of the Supplementary Material

(Hou et al., 2026) by introducing the L_1 constrained score space. The HAL-MLE, f_n, has one L_1 norm constraint on the coefficients, and thus solves $J_n - 1$ score equations. We name the linear span of these $J_n - 1$ scores as the L_1 constrained score space, and this space is a closed linear subspace of $D^k(\mathcal{R}_n)$.

Let $\{\phi_j^* : j \in \mathcal{R}_{n,0}\}$ be an orthonormal basis of $\operatorname{span}\{\phi_j : j \in \mathcal{R}_{n,0}\}$ under the inner product $\langle f, g\rangle_{\beta_{n,0}}$. We define a normalized score vector at x by

$$\bar{\phi}_{n,x}(X) \equiv J_{n,0}^{-1/2} \sum_{j\in\mathcal{R}_{n,0}} \phi_j^*(x)\phi_j^*(X).$$

For each value $x \in [0,1]$, $\bar{\phi}_{n,x}$ is a function on $[0,1]$. Also,

$$J_{n,0}^{-1/2}\bar{\phi}_{n,x}(X) = J_{n,0}^{-1} \sum_{j\in\mathcal{R}_{n,0}} \phi_j^*(x)\phi_j^*(X),$$

which is a linear combination of the orthonormal basis functions ϕ_j^*, and we believe it to be uniformly bounded. Additionally, we use notation $O^+(r(n))$ if the term is $O(r(n)(\log n)^p)$ for some finite p, and similarly we define $O_p^+(r(n))$. Note that p can be negative, and usually we can determine this p based on undersmoothening, and it does not influence the main rate $r(n)$.

THEOREM 4.4 (Asymptotic Linearity of HAL-MLE). *Let $J_{n,0}$ satisfy $J_{n,0}^{-(k+1)} = o((J_{n,0}/n)^{1/2})$. Suppose the following conditions hold for an integer $k \geq 1$:*

1. *(Basis boundedness) $J_{n,0}^{-1/2}\|\bar{\phi}_{n,x}\|_\infty = O(1)$;*
2. *(Smoothness) $f_0 \in D_U^{(k)}([0,1])$ for some $U < \infty$;*
3. *(Uniform Approximation of $D_U^{(k)}([0,1])$ by $D^{(k)}(\mathcal{R}_{n,0})$)*
$$\sup_{f\in D_U^{(k)}([0,1])} \inf_{g\in D^{(k)}(\mathcal{R}_{n,0})} \|f-g\|_\infty = O^+\big(J_{n,0}^{-(k+1)}\big).$$
4. *($L^2(\mu)$ Approximation of $D_U^{(k)}([0,1])$ by $D^{(k)}(\mathcal{R}_n)$)*
$$\sup_{f\in D_U^{(k)}([0,1])} \inf_{g\in D^{(k)}(\mathcal{R}_n)} \|f-g\|_{L^2(\mu)} = O^+\big(J_n^{-(k+1)}\big),$$
and can be extended to the corresponding L_1 constrained score space in $D^{(k)}(\mathcal{R}_n)$.

Then, with cross-validated or slightly undersmoothened f_n, for each fixed $x \in [0,1]$,

$$\sqrt{\frac{n}{J_{n,0}}}\big(f_n(x) - f_0(x)\big) = \sqrt{n}\,(P_n - P_0)\big[S_{f_{n,0}}(\bar{\phi}_{n,x})\big] + o_P(1).$$

The latter term equals $n^{1/2}$-scaled empirical mean of independent mean zero random variables $S_{f_{n,0}}(\bar{\phi}_{n,x})$ with finite variance. $\sqrt{J_{n,0}}S_{f_{n,0}}(\bar{\phi}_{n,x})$ serves as the influence curve of the HAL-MLE $f_n(x)$ considered as an estimator of the oracle $f_{n,0}(x)$.

Here we explain the plausibility and understanding for these approximation assumptions. The bias is identified as the uniform approximation error between the finite-dimensional working model $D^{(k)}(\mathcal{R}_{n,0})$ and $D_U^{(k)}([0,1])$. van der Laan (2023, Appendices D and F) show that such a set $\mathcal{R}_{n,0}$ of size $J_{n,0}$ exists for all sizes $J_{n,0}$. Therefore, we could simply select $\mathcal{R}_{n,0}$ as such a set and choose it for $J_{n,0} = J_n$. Moreover, in this article it is argued that, due to HAL achieving the L^2 rate of convergence $O_p(n^{-(k+1)/(2k+3)})$, the sets of non-zero coefficients in the HAL fit should satisfy an adaptive version of this approximation condition allowing this same approximation error w.r.t. true target function, and that under some undersmoothing it will satisfy the (non-adaptive) uniform approximation condition. And the L^2 approximation error is a weaker condition on $D^{(k)}(\mathcal{R}_n)$. Its extension to the L_1 constrained score space can be explained by approximating one basis function by other basis functions in the score space.

REMARK 4.5 (The Necessity for Cross-Validation and Undersmoothing). The L_1-norm in the HAL-MLE drives the number of basis functions in $\mathcal{R}_n$. This number J_n needs to balance between the bias $O^+\left(J_n^{-(k+1)}\right)$ of the working model $D^{(k)}(\mathcal{R}_n)$ and variance $\sqrt{\frac{n}{J_n}}$ or the MLE for this working model. Cross-validation can guarantee the optimal choice (Van Der Laan and Dudoit, 2003; Vaart, Dudoit and Laan, 2006). One may need some undersmoothing so that the bias is reduced by a $\log n$ factor, making it negligible relative to the standard error.

THEOREM 4.6 (Pointwise Asymptotic Normality for HAL-MLE). *Under the conditions of Theorem 4.4, let $\sigma_{n,0}^2(x) = P_0(\bar{\phi}_{n,x} - P_{\beta_{n,0}}\bar{\phi}_{n,x})^2$, and then we have*

$$\sigma_{n,0}^{-1}\left(\frac{n}{J_{n,0}}\right)^{1/2}\left(f_n - f_0\right)(x) \Rightarrow_d N(0,1). \tag{4.3}$$

A natural question is whether we can extend these to the corresponding density estimation. Notice that what we demonstrated is the pointwise asymptotic normality instead of the weak convergence of $(f_n - f_0)$ in function space. Thus, we cannot even apply the functional delta method. However, we construct the proof of the following corollary by recognizing that it is driven entirely by $f_n - f_0$, while the normalizing constant is a pathwise differentiable plug-in estimator with no contribution to the linearization.

COROLLARY 4.7 (Delta-method for Density Estimation). *Under the conditions of Theorem 4.4, let $g(f)(x) = \log p_f(x)$, then we have,*

$$(n/J_{n,0})^{1/2}(g(f_n)(x) - g(f_0)(x)) = (n/J_{n,0})^{1/2}(f_n(x) - f_0(x)) + o_P(1).$$

Therefore, $\log p_{f_n}(x) - \log p_{f_0}(x)$ behaves as $(f_n - f_0)(x)$, which proves that

$$(n/J_{n,0})^{1/2}(p_{f_n} - p_{f_0})(x) = p_0(x)(n/J_{n,0})^{1/2}(f_n - f_0)(x) + o_P(1).$$

This extends the pointwise asymptotic normality and the uniform convergence from f_n to p_{f_n}.

4.3. *Uniform Convergence for HAL-MLE.* The uniform convergence of HAL-MLE is a direct consequence of the pointwise asymptotic normality.

THEOREM 4.8 (Uniform Convergence for HAL-MLE). *Extending the Basis Boundedness Assumption to hold uniformly for $x \in [0,1]$ a.s., while maintaining the other assumptions of Theorem 4.4, then*

$$\sup_{x\in[0,1]} \sqrt{n/J_{n,0}}\,|f_n(x) - f_0(x)| = O_P(\sqrt{\log n}).$$

By selecting $J_{n,0} \asymp n^{1/(2k+3)}$ up till $\log n$-factors, it follows

$$\|f_n - f_0\|_\infty = O_P^+\left(n^{-(k+1)/(2k+3)}\right).$$

REMARK 4.9 (Zero-order Case). When $k = 0$, the uniform rate above need not hold, but one still has $\|f_n - f_0\|_\infty = o_P(1)$ based on van der Laan and Bibaut (2017).

In Remark 4.2, we mentioned the difference between the log-spline convergence and HAL-MLE convergence. Here we provide more details. First, in Stone (1990), they deal with a parametric MLE on uniform knots, whereas we deal with data-adaptive knots. Second, the convergence they showed is between the estimation and the parametric oracle MLE. Third, their uniform convergence rate relies on the assumption of the size of the post-selection working model to be $o(n^{\frac{1}{2}-\epsilon})$ for some $\epsilon > 0$; however, there is no theory suggesting that AIC or BIC can select the right size of the post-selection working model, unlike the theoretical guarantee in Van Der Laan and Dudoit (2003) for cross-validation. Here, HAL-MLE shows a stronger result considering the bias between the oracle and the truth with the simplest setting of just using the data-adaptive knot points and cross-validation.

4.4. *Generalizability of the Proofs.* Another interesting discussion is the generalizability of these proofs to different basis functions but asymptotically spanning the same function space, such as the B-splines and falling factorial basis. For the B-splines, they are linear combinations of the truncated power basis (De Boor, 1976). For the falling factorial basis, they are identical to the truncated power basis up to a constant when $k = 0$ and $k = 1$. When $k \geq 2$, the proximity of the truncated power basis and the falling factorial basis is discussed in Tibshirani et al. (2022, Chapter 10.1). Nevertheless, when $k \geq 1$, the falling factorial basis is continuous and therefore trivally càdlàg. Thus, these three basis systems should span the same space. The key assumptions of HAL-MLE proofs rely on i) bounding the SVN, ii) being an MLE on a fine enough working model. This is the reason why we have the same L^2 convergence rate as in the regression case in Fang, Guntuboyina and Sen (2021); Ki, Fang and Guntuboyina (2024), even though the multivariate basis system is different. So our guess is that these proofs can be generalized to the B-splines and falling factorial basis under the same setting. However, the representation of TV is complicated for the B-splines with data-adaptive knots as mentioned in Wang, Smola and Tibshirani (2014, Section 3).

4.5. *Variance Estimation for Density.* Based on pointwise asymptotic normality, we propose a variance estimator using the Delta method, analogous to the variance estimator for generalized linear models (GLMs). Consider the parametric working model for HAL-MLE, i.e.,

$$f_n(x) = \phi(x)^\top \beta_n,$$

where ϕ denotes the vector of basis functions in $\mathcal{R}_n$. Consider this working model directly as a parametric model for the density itself with a truth β_0, we have a finite-dimensional parameter β_n, and $\beta_n \xrightarrow{P} \beta_0$. The density is then parameterized by β_n, and we denote $p_{f_n}(x) = p_{\beta_n}(x)$. Using the Delta method, we get the first-order Taylor expansion of $p_{f_n}(x)$ around β_n,

$$p_{f_n}(x) = p_{\beta_n}(x) = p_{\beta_0}(x) + \nabla_\beta p_{\beta_n}(x)(\beta_n - \beta_0) + o_P(\|\beta_n - \beta_0\|),$$

where $\nabla_\beta p_{\beta_n}(x)$ is the gradient of $p_{\beta_n}(x)$ with respect to β_n. Ignoring the second-order term, and taking the variance of both sides, we get

$$\mathrm{Var}(p_{f_n}(x)) = \mathrm{Var}(p_{\beta_n}(x) - p_{\beta_0}(x)) \approx \mathrm{Var}(\nabla_\beta p_{\beta_n}(x)(\beta_n - \beta_0)).$$

Denoting $\mathrm{Var}(p_{\beta_n}(x))$ as $\sigma_n^2(x)$, we obtain a plug-in estimator of the variance $\sigma_{n,0}^2(x)$ defined in Theorem 4.6, as follows:

$$\begin{aligned} \sigma_n^2(x) =& (\nabla_\beta p_{\beta_n}(x))^\top \, \widehat{\mathrm{Cov}}(\beta_n) \, (\nabla_\beta p_{\beta_n}(x)) \\ =& (p_{\beta_n}(x))^2 \left[S_{f_n}(\phi)(x) \right]^\top \widehat{\mathrm{Cov}}(\beta_n) \left[S_{f_n}(\phi)(x) \right]. \end{aligned} \tag{4.4}$$

And the confidence interval can be constructed accordingly based on the z-statistic. The above formula $\sigma^2_{n,0}$ in Theorem 4.6 for the asymptotic variance relies on the orthonormal basis formulation, which makes it not practical to estimate it. This delta-method formula is a standard sandwich variance estimator for a function of the parameters of the working model. The covariance matrix of β_n is estimated with the empirical covariance of the inverse of the information matrix applied to the score vector $(S_{f_n}(\phi)$, where $\phi \equiv (\phi_1, \cdots, \phi_{J_n})^\top$, corresponds to robust estimation of the covariance matrix of the MLE for a misspecified parametric working model. In our case, note that this observed information matrix actually also equals the covariance of the scores under P_{f_n}: $I_n(\beta_n)(j_1, j_2) = P_{f_n} S_{f_n}(\phi_{j_1}) S_{f_n}(\phi_{j_2})$. Since our working model $D^{(k)}(\mathcal{R}_n)$ will approximate the true f_0, we could also replace $I_n(\beta_n)$ by $I_{n,emp(\beta_n)} = P_n S_{f_n}(\phi) S_{f_n}(\phi)^\top$. We claim that the latter is more robust than the one using $I_n(\beta_n)$ and is thus recommended.

4.5.1. *Covariance Estimation for β_n.* Our delta-method variance estimator relies on an inverse of the information matrix, such as its empirical version

$$I_{n,emp}(\beta_n) = \frac{1}{n} \sum_{i=1}^{n} S_{f_n}(\phi)(x_i)\, S_{f_n}(\phi)(x_i)^\top.$$

The inverse can become unstable when some of the basis functions are highly correlated. For numerical stability, we might add a ridge regularization to the diagonal of $I_n(\beta_n)$ before inverting, in case it is nearly singular. Following the uniform convergence theory of the HAL, the convergence rate is $\sqrt{\frac{J_{n,0}}{n}}$. Thus, the $\sigma^2_n(x) = O_p(\frac{J_{n,0}}{n})$, and this implies $I_n(\beta_n)^{-1} = O_p(J_{n,0})$. So, its minimum eigenvalue should be of order $\frac{1}{J_{n,0}}$.

For the first order HAL, $J_{n,0} \asymp^{+} (n^{\frac{1}{5}})$. Thus, we should add a regularization parameter at least smaller than $O_p(n^{-\frac{1}{5}})$. On the other hand, this delta-method variance estimator ignores the bias of the working model so that some overestimation of this variance would be welcome for better coverage. Overall, the robust selection of such a constant is an interesting problem to be addressed in future work. In our practical simulations, we added a regularization parameter of size $O_p(n^{-\frac{1}{5}})$.

5. HAL-MLE for Pathwise Differentiable Statistical Estimands. One could use the density estimator for statistical estimands such as moments, the median, the CDF, or the survival probability. Let $\Psi : \mathcal{M} \to \mathbb{R}$ be a pathwise differentiable target parameter with canonical gradient (also called efficient influence curve) D^*_P at P where $\mathcal{M} = \{P_f : f \in D^{(k)}_U([0,1]^d\}$, then, for a path P^h_ε through P with direction h, $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Psi(P^h_\varepsilon) = \mathbb{E}_P[D^*_P\, h]$. Let $R(P, P_0) \equiv \Psi(P) - \Psi(P_0) + P_0 D^*_P$ be the exact remainder implied by the canonical gradient, which is known to be a second-order difference due to it being the exact remainder in a first order Tailor expansion $R(P, P_0) = \Psi(P) - \Psi(P_0) - (P - P_0) D^*_P$. We have $\Psi(P) = \Psi(P_f) = \Psi^F(f)$ for a $\Psi^F : D^{(k)}_U([0,1]^d) \to \mathbb{R}$. We can represent $D^*_P = D^*_f$ with $f = f(P)$.

5.1. *"Naive" Approach: Plug-in.* Plugging in the density would be a natural approach, and we will show that it yields an asymptotically efficient estimator, possibly achieved by using undersmoothing. We here provide a standard TMLE-type analysis establishing asymptotic efficiency under the assumption that $D^{(k)}(\mathcal{R}_n)$ approximates $D^{(k)}_U([0,1])$.

The tangent space generated by paths through f is given by closure of linear span $\{S_f(\phi) : \phi \in D^{(k)}_U([0,1])\}$. Thus, we can represent $D^*_f = S_f(\phi_f)$ for some $\phi_f \in D^{(k)}_M([0,1])$ for $\forall k$ including 0. Therefore, if the basis $\mathcal{R}_n$ is chosen so that $D^{(k)}(\mathcal{R}_n)$ yields a $O^+(1/J^{k+1}_n)$ L^2

approximation of $D^{(k)}_U([0,1])$, then we can find a projection $\tilde{\phi}_f$ of ϕ_f onto $D^{(k)}(\mathcal{R}_n)$ that approximates ϕ_f in L^2-norm within distance $O^+(1/J_n^{k+1})$.

For simplicity, firstly, consider the case that f_n is a relaxed HAL-MLE so that it equals an MLE over $D^{(k)}(\mathcal{R}_n)$, while we assume that it has been established that $\|\beta_n\| = o_p(1)$ as it would be for the HAL-MLE. Then, $P_n S_{f_n}(\phi_j) = 0$ for all $j \in \mathcal{R}_n$. Let $\tilde{D}_{f_n}$ be the projection in $L^2(\mu)$ of $D^*_{f_n}$ onto $\{S_{f_n}(\phi_j) : j \in \mathcal{R}_n\}$. Then, $P_n \tilde{D}_{f_n} = 0$. In that case if $\|\tilde{D}_{f_n} - D^*_{f_n}\|_\mu = o_p^+(1/J_n^{k+1})$, then it follows that $P_n D^*_{f_n} = o_p(n^{-1/2})$ and the efficiency theorem below applies. For the HAL-MLE we already know we control $\|\beta_n\|_1 = O(1)$, but we would need to show that the scores $P_n S_{f_n}(\phi_j)$, $j \in \mathcal{R}_n$ are solved good enough so that $P_n \tilde{D}_{f_n} = o_p(n^{-1/2})$. In Section S.6 of the Supplementary Material (Hou et al., 2026), we show that this can generally be arranged by extending the approximation condition to the L_1 constrained score space.

THEOREM 5.1 (Efficient Influence Curve Approximation). *Suppose the sieve $D^{(k)}(\mathcal{R}_n)$ satisfies the L^2-approximation and it can be extended to its corresponding L_1 constrained score space, and maintain the Smoothness Assumptions in Theorem 4.4. Let f_n be the cross-validated HAL–MLE over $D^{(k)}(\mathcal{R}_n)$. Then*

$$P_n\, D^*_{f_n} = o_p\big(n^{-1/2}\big).$$

This directly leads to the asymptotic efficiency theorem.

THEOREM 5.2 (Asymptotic Efficiency of Plug-in HAL-MLE). *Suppose the following conditions hold:*

1. *(L^2-Approximation) $D^{(k)}(\mathcal{R}_n)$ satisfies the L^2-approximation and it can be extended to its corresponding L_1 constrained score space.*
2. *(Negligible Remainder) $R(P_{f_n}, P_0) = o_p\big(n^{-1/2}\big)$.*
3. *($L^2(P_0)$-continuity of EIC) $P_0\{(D^*_{f_n} - D^*_{f_0})^2\} = o_p\big(1\big)$.*

Then the plug-in estimator $\Psi^F(f_n)$ satisfies

$$\Psi^F(f_n) - \Psi^F(f_0) = (P_n - P_0)\, D^*_{f_0} \;+\; o_p\big(n^{-1/2}\big),$$

and is therefore asymptotically efficient.

We discussed the plausibility of the first assumption in Section 4.2. And here we demonstrate that the second and third assumptions are not only reasonable but also very weak with the examples of survival probability and moments. We leave the argument of median, which is slightly complicated, to Section S.6 of the Supplementary Material (Hou et al., 2026). For survival probability at x_0, we have

$$P_0(D^*_{f_n} - D^*_{f_0}) = \int I(x > x_0)(p_{f_n} - p_{f_0})\mathrm{d}x = O(1)o_p(\|p_{f_n} - p_{f_0}\|_\mu) = o_p(n^{-\frac{k+1}{2k+3}}),$$

by the Cauchy-Schwartz inequality. This implies $P_0\{(D^*_{f_n} - D^*_{f_0})^2\} = o_p(n^{-\frac{2k+2}{2k+3}}) = o_p\big(1\big)$.

$$R(P_{f_n}, P_0) = S_{f_n}(x_0) - S_0(x_0) + P_0(I(x > x_0) - P_{f_n}(x > x_0)) = 0.$$

The argument is basically the same for moments,

$$P_0(D^*_{f_n} - D^*_{f_0}) = \int x^k(p_{f_n} - p_{f_0})\mathrm{d}x = o_p(n^{-\frac{k+1}{2k+3}}),$$

by Cauchy-Schwartz inequality.

$$R(P_{f_n}, P_0) = \mathbb{E}_{P_{f_n}}(x^k) - \mathbb{E}_{P_0}(x^k) + P_0(x^k - \mathbb{E}_{P_{f_n}}(x^k)) = 0.$$

5.2. *HAL-TMLE.* We can exploit the information about the target parameter more effectively by incorporating the TMLE framework, namely HAL-TMLE. We found that the targeting step is very compatible with the link function, and each target parameter corresponds to a different targeting.

Targeting Step: We construct a one-dimensional submodel $\{p_{f_\varepsilon^h} : \varepsilon\}$ indexed by a direction h of a path $\{f_\varepsilon^h : \varepsilon\}$ through f that satisfies

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} \log p_{f_\varepsilon^h}\bigg|_{\varepsilon=0} = D_f^*(x), \tag{5.1}$$

MLE Step: Specifically, we select $f_\varepsilon^h = f + \varepsilon h$, and thus we have

$$p_{f_\varepsilon^h} = \frac{\exp\big(f + \varepsilon h\big)}{\displaystyle\int \exp\big(f + \varepsilon h\big)\mathrm{d}x},$$

where f plays the role of $f_{n,\beta_n(M)}$. We can simply select $h = h_f = D_f^*$ to make this path a local least favorable path (LLFP). Let's denote the resulting path with $f_{n,\varepsilon}$ Then, the first step TMLE update would be defined by f_{n,ε_n^1} with $\varepsilon_n^1 = \arg\min_\varepsilon -P_n \log p_{f_{n,\varepsilon}}$. One can set $f_n^1 = f_{n,\varepsilon_n^1}$ and define $f_{n,\varepsilon}^1$ as this LLFP but now with off-set f_n^1, and define $\varepsilon_n^2 = \arg\min_\varepsilon -P_n \log p_{f^1_{n,\varepsilon_n^2}}$. We can iterate this TMLE-updating till $P_n D^*_{f_n^k} \leq \sigma_n/(n^{1/2}\log n)$, where $\sigma_n^2 = (P_n D^*_{f_n})^2$ is the estimator of the sample variance of the efficient influence curve at the initial estimator. The final update satisfying this criterion is then the iterative TMLE f_n^*, implying the plug-in TMLE $\Psi^F(f_n^*)$ of $\Psi^F(f_0)$.

A theoretical concern in iterative TMLE is the preservation of the function class. The HAL-MLE f_n belongs to a function space whose induced distribution $P^0_{f_n}$ lies in a Donsker class with covering number equivalent to that of $D^{(k)}([0,1])$. When applying a finite number of targeting steps, this property is preserved. However, performing infinitely many TMLE iterations may drive the estimator outside of this Donsker class.

According to van der Laan (2017), a single targeting step already suffices, because the initial HAL-MLE f_n converges to f_0 at a rate faster than $n^{-1/4}$ and we also satisfy the positivity condition, the single-step update f_n^1 obtained by fluctuating along the least favorable submodel satisfies $P_n D^*_{f_n^1} = o_p(n^{-1/2})$. Consequently, f_n^1 also attains the same asymptotic efficiency, while preserving the convergence rate and sectional variation norm of the initial HAL-MLE. We formally state the theorem as follows.

THEOREM 5.3 (Asymptotic Efficiency of Single-step HAL-TMLE). *Suppose we impose a single targeting step to f_n and yield f_n^1. And suppose that the second and third assumptions hold for f_n^1, that is,*

1. *(Negligible Remainder)* $R(P_{f_n^1}, P_0) = o_p\big(n^{-1/2}\big)$.
2. *($L^2(P_0)$-continuity of EIC)* $P_0\{(D^*_{f_n^1} - D^*_{f_0})^2\} = o_p\big(1\big)$.

Then $P_n D^*_{f_n^1} = o_p(n^{-1/2})$, and thus the single-step HAL-TMLE is asymptotically efficient.

Here are some examples for which we can compute this TMLE:

EXAMPLE 5.4 (Survival function and CDF). For the marginal survival function $\Psi^F(f) = \int_{x_0}^1 p_f(y)dy$ at x_0, we have $D_f^*(x) = I(x > x_0) - \mathbb{E}_{P_f}\big[I(x > x_0)\big]$. We can thus select $h = I(x > x_0)$. Note that, technically, we need $\mathbb{E}_{P_{f_n}}[h] = 0$, but the constant will cancel out by the normalizing constant. Similarly, for CDF F_f at x_0 we can select $h = I(x \leq x_0)$.

EXAMPLE 5.5 (Median). For the median $\Psi^F(f) = F_f^{-1}(0.5)$ we have $D_f^*(x) = \frac{\frac{1}{2} - I(x < F_f^{-1}(0.5))}{f(F_f^{-1}(0.5))}$, and thus $h = I(X < F_f^{-1}(0.5))$. We need to estimate the median based on the initial and then do the targeting. Note that the targeting of median is trivially generalized to any percentile. Since the density estimation changes from f_n to f_n^k after the k steps targeting, we can iteratively do the targeting of the median. However, in Theorem 5.3, we show that a single step targeting already guarantees $P_n D^*_{f_n^1} = o_p(n^{-1/2})$, and thus the asymptotic efficiency holds.

EXAMPLE 5.6 (k-th order moments). For the k-th order moment $\Psi^F(f) = \int x^k p_f(x)\mathrm{d}x$ we have $D_f^*(x) = x^k - \Psi^F(f)$, and thus $h = x^k$. Note that this is just the parametric basis. This implies a motivation for not penalizing the parametric part, like LAS and the classic TF, in the sense that the resulting HAL fit would be targeting all its moments.

Note that the LLFPs for survival probability and moments are actually their universal least favorable paths (ULFP), and their corresponding single-step HAL-TMLE is the one-step HAL-TMLE in van der Laan (2017); van der Laan and Gruber (2016); Van der Laan and Rose (2018), and thus we have $P_n D^*_{f_n^1} = 0$ exactly. Also, by van der Laan (2017), f_n^1 and f_n preserve the same convergence rate to f_0. Thus, we can extend the arguments of the assumptions of the plug-in HAL-MLE to the single-step HAL-TMLE.

5.3. *Theoretical Property and Variance Estimation.* As established in Section 5.1 and Section 5.2, the score equation is solved at least at level $o_p(n^{-1/2})$. Thus, both estimators are asymptotically efficient and asymptotically linear, with their proofs unified in Section S.6 of the Supplementary Material (Hou et al., 2026). Since TMLE is asymptotically linear with influence curve $D^*_{f_n^1}$, variance estimation for $\Psi^F(f_n)$ can be obtained directly by computing the sample variance of the estimated $D^*_{f_n^1}(x_i)$.

6. Simulation. This section documents the Monte Carlo simulation results for the HAL-MLE. We (i) test the uniform convergence and pointwise asymptotic normality of the HAL-MLE, (ii) test the asymptotic efficiency of the plug-in HAL-MLE and HAL-TMLE, and (iii) compare it with the logspline, TF and its equivalent variant with penalty on the parametric part (TFPP), and KDE methods, over a vast range of different DGPs and sample sizes.

6.1. *DGPs.* We use six univariate densities on $[0, 1]$ across continuous vs discrete, concentrated vs heavy-tailed, and smooth vs oscillatory (single-scale vs multi-scale). The six DGPs are: Truncated Normal (TN), Truncated GMM with three equally spaced modes (GS3), Truncated GMM with three modes of differing scales and weights (GA3), Truncated GMM with five spikes and background (GS5), Step Function (Step), and Sinusoidal (Sine), as shown in Figure 1. We provide the ground–truth density p_0 and population parameters, such as moments and medians, in Section S.7 of the Supplementary Material (Hou et al., 2026).

6.2. *HAL-MLE density estimation.* We here evaluate the uniform convergence and pointwise asymptotic normality of the HAL-MLE.

6.2.1. *Uniform convergence.* We consider eight sample sizes, from $n = 25$ to $n = 3200$, with 1000 independent random samples per DGP at each n. For each sample, we approximate the sup–norm error $\|p_{f_n} - p_0\|_\infty$ by evaluating both the estimator and the truth on a uniform grid of 201 points over $[0, 1]$ and taking the maximum absolute deviation over grid points. By Theorem 4.8, the uniform convergence rate is $O_p^+\left(n^{-(k+1)/(2k+3)}\right)$ for basis order $k \in \{1, 2\}$.

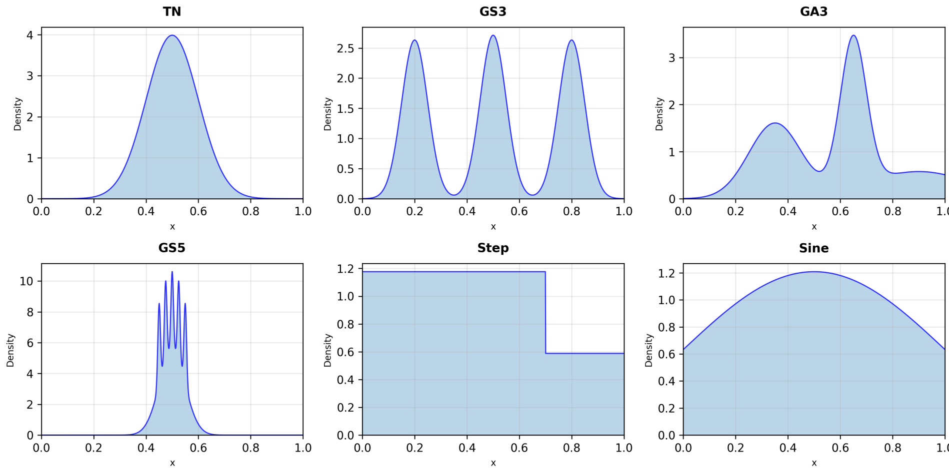

FIG 1. *Six DGPs: (a) TN, (b) GS3, (c) GA3, (d) GS5, (e) Step, (f) Sine.*

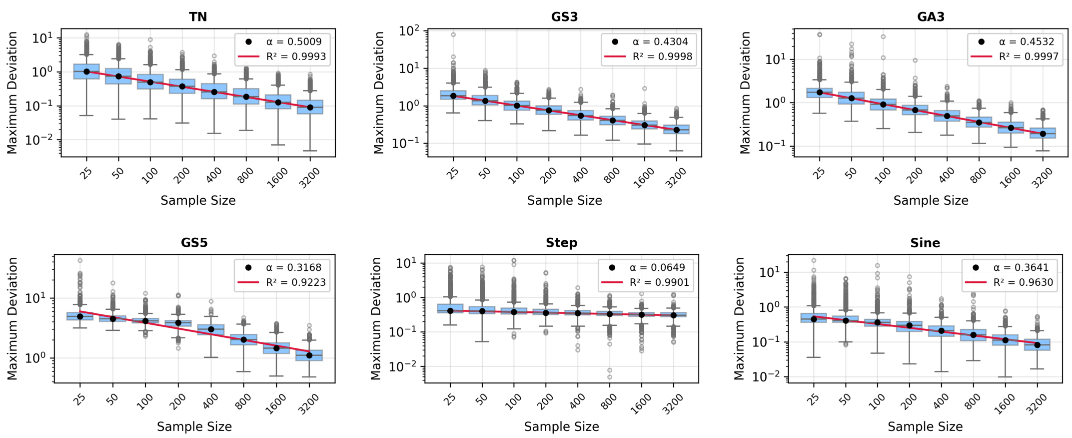

FIG 2. *Uniform–convergence sup–norm error (log–scale) by DGP. Panels (row–wise, left to right): (a) TN, (b) GS3, (c) GA3, (d) GS5, (e) Step, (f) Sine.*

Metric	TN	GS3	GA3	GS5	Step	Sine
α (Decay Exponent)	0.5009	0.4304	0.4532	0.3168	0.0649	0.3641
R^2 (Goodness of Fit)	0.9993	0.9998	0.9997	0.9223	0.9901	0.9630
Significant ($p < 0.05$)	Yes	Yes	Yes	Yes	Yes	Yes

TABLE 1

Uniform convergence scaling analysis. α represents the decay exponent in the power law error $\propto n^{-\alpha}$. R^2 measures goodness of fit on the log-log scale. Significance tests H_0 : slope $= 0$ vs H_1 : slope $\neq 0$ at $\alpha = 0.05$.

Because k is selected by cross-validation over this set, we expect at least $O_p\left(n^{-1/3}\right)$ decay, even though we do not have the uniform convergence rate for $k = 0$. This can be read off from the median lines of the boxplots below as n increases. We aggregate errors across samples and summarize their distributions by sample size as shown in Figure 2.

We regress the median error on the sample size on a log-log scale and report the slope and R^2 in Table 1. The results are shown below in Table 1.

DGP	N=25	N=50	N=100	N=200	N=400	N=800	N=1600	N=3200
TN	(94.7, 93.6)	(93.8, 92.9)	(93.7, 93.2)	(93.7, 92.1)	(92.6, 92.2)	(95.1, 92.1)	(95.5, 92.2)	(91.2, 92.0)
GS3	(83.7, 90.9)	(89.9, 91.2)	(90.0, 90.8)	(90.5, 91.3)	(91.2, 91.1)	(93.3, 91.4)	(92.2, 91.4)	(91.7, 91.4)
GA3	(81.5, 91.9)	(85.2, 92.4)	(87.7, 92.6)	(88.6, 92.9)	(90.9, 92.9)	(94.9, 92.8)	(95.6, 92.6)	(95.2, 92.8)
GS5	(93.8, 95.4)	(91.2, 94.1)	(88.9, 92.7)	(90.4, 93.7)	(95.3, 96.5)	(95.6, 96.5)	(94.5, 96.6)	(94.7, 96.4)
Step	(90.0, 94.2)	(85.9, 92.2)	(81.9, 90.6)	(80.3, 90.3)	(82.9, 90.0)	(90.0, 90.1)	(93.2, 90.5)	(94.2, 91.0)
Sine	(92.5, 94.3)	(91.0, 94.4)	(88.5, 94.7)	(85.5, 94.1)	(89.5, 92.8)	(94.2, 93.5)	(95.9, 93.5)	(96.2, 93.5)

TABLE 2

Coverage probabilities (%) for 95% confidence intervals across DGPs and sample sizes.

This Table 1 verifies that, if there is a convergence rate, the median line should be approximately a straight line with a decreasing slope, expected to be around or above $-\frac{1}{3}$. The Step Function might be too easy and does not have a huge space for improvement as the sample size increases.

6.2.2. *Pointwise Asymptotic Normality: pointwise SE, CI width, and coverage.* We assess the pointwise asymptotic normality of the HAL-MLE density mentioned in Theorem 4.6 and evaluate the performance of density variance estimation via the delta-method in Eq. 4.4. We assess the density variance estimation performance by introducing the oracle sd, which is the empirical standard deviation of the fitted densities across 1000 samples. We construct the CI by the z-statistic of the HAL-MLE point estimate. Then, we evaluate the variance estimator by the coverage of the CI using the oracle coverage as the benchmark.

For each DGP and sample size n, we evaluate on the 201 grid points between the first and last observation and summarize (i) interval widths and (ii) coverage of the truth $p_0(x)$. Coverage is computed pointwise as the fraction of replicates whose 95% CI contains the truth at each grid point. The coverage of the Delta-method variance estimator is the mean coverage over the 201 grid points. The full results, including the CI widths and the coverage of each grid point, are provided in Section S.9.1 of the Supplementary Material (Hou et al., 2026). Table 2 reports mean coverage summaries by DGP and sample size, shown as (plug–in, oracle). Each cell shows (estimated coverage, oracle coverage).

The oracle CI coverage is always around 95%, which means the density estimation is pointwise asymptotically normal. For most of the DGPs, the Delta-method variance estimator is also very close to the oracle CI coverage. A limitation of the Delta-method is that it does not work as well on the extrapolated region, where we have no data, even though the density estimation is still valid in the sense of pointwise asymptotic normality.

6.2.3. *Asymptotic efficiency.* We here evaluate the asymptotic efficiency of the plug-in HAL-MLE and HAL-TMLE by comparing their performance in estimating some statistical estimands such as the mean, median, survival at 0.5, and second moment, against their corresponding asymptotically efficient estimators, i.e., the sample mean, etc.

Across DGPs and sample sizes, we compare the bias, variance, MSE, and the ratio bias/sd of plug-in HAL-MLE and HAL-TMLE with the asymptotically efficient estimators. The results of GA3 are shown below in Figure 3. The full results are provided in Section S.9.2 of the Supplementary Material (Hou et al., 2026).

They are comparable and thus verify the asymptotic efficiency of the plug-in HAL-MLE and HAL-TMLE. We can see that sometimes the plug-in HAL-MLE has super-efficiency over the asymptotically efficient estimator when the sample size is small. But after targeting, the EIC equation is solved, and the HAL-TMLE performance is the same as the asymptotically efficient estimator uptill some numerical errors. In the univariate case, we cannot expect the performance of Plug-in HAL-MLE and HAL-TMLE to be significantly better than their

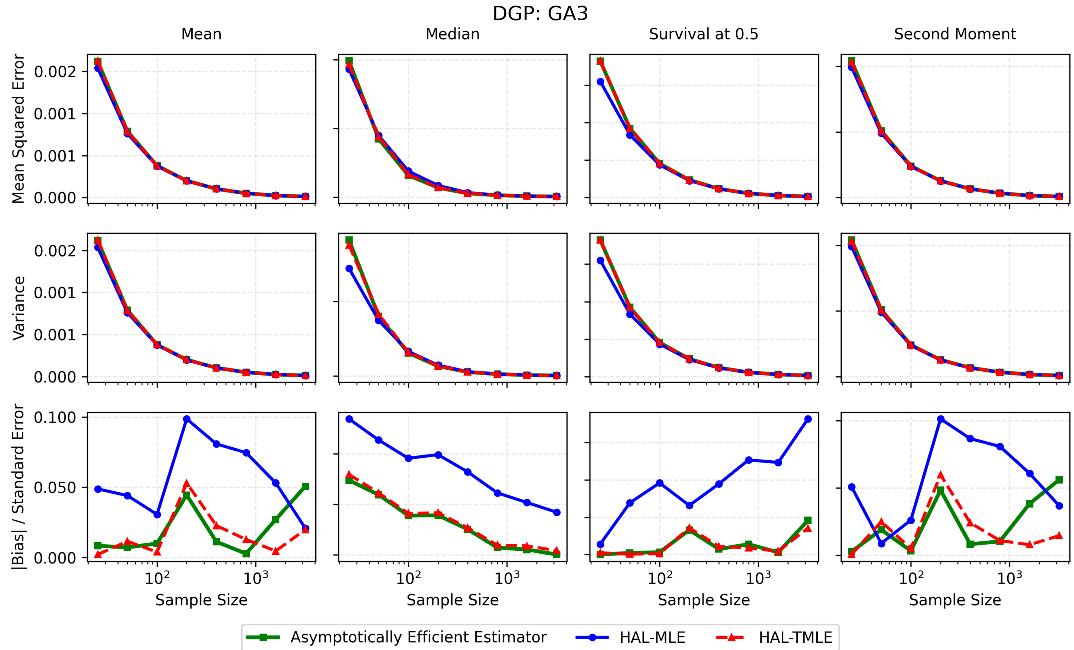

FIG 3. *Representative asymptotic–efficiency comparison (DGP: GA3). Columns show four statistical estimands (Mean, Median, Survival at 0.5, Second Moment) and rows show three metrics (MSE, Variance, |Bias|/SE). Curves compare the asymptotically efficient estimator (green), HAL–MLE (blue), and HAL–TMLE (red).*

corresponding asymptotic efficient estimators, but one major advantage is that we can provide nonparametric inference using the EIC-based confidence interval.

6.3. *EIC–based variance estimation for statistical estimands.* We evaluate the EIC-based variance estimation introduced in Section 5.3 for the statistical estimands by computing the plug-in standard errors via the efficient influence curve using the oracle CI coverage for statistical estimands as the benchmark. Also, intuitively, another variance estimator could be proposed by applying the delta method to the density again. However, in practice we find the EIC-based variance estimator is more robust and has a better coverage than the delta method variance estimator. For all statistical estimands, the EIC-based variance estimator and the oracle coverage are all very close to 95%. The full figure panels of CI width and coverage for each DGP are provided in Section S.9.3 of the Supplementary Material (Hou et al., 2026).

6.4. *Comparison with Existing Methods.* We compare the HAL-MLE with the existing methods, including the TF, TFPP, logspline, and KDE methods. TF(PP) are carried out here with `CVXPY` as well for the sake of fairness. However, we also implemented an ADMM solver following Ramdas and Tibshirani (2016) for TF on the same uniform grid, with similar performance to the `CVXPY` implementation; see Section S.8 of the Supplementary Material (Hou et al., 2026). We evaluate the performance of the existing methods at $n = 800$ over a common grid of 19 points on $[0.05, 0.95]$; and report the MSE in Figure 4.

6.4.1. *Hyperparameter selection.* The hyperparameter selection for the TF(PP), logspline, and KDE methods are as follows: (i) The TF and TFPP methods have a λ in $[10^{-3}, 10^{6}]$ for the ℓ_1 penalty and k for the kth order finite differences ($k \in \{0, 1, 2\}$). (ii) The logspline method is a spline basis for the log-density with default hyperparameter choices in the R package

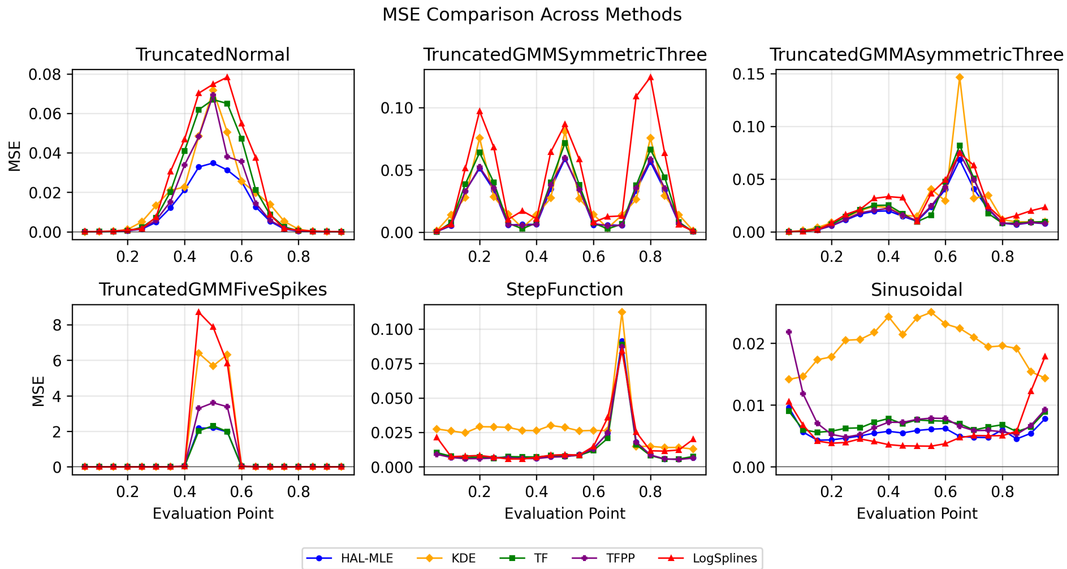

FIG 4. *MSE at $n = 800$ across six DGPs (columns) for five estimators (legend).*

`logspline`. (iii) The KDE method is a kernel smoother with jointly tuned kernel (Gaussian, Epanechnikov, tophat) and bandwidth h in $[10^{-3}, 10^{6}]$.

Some observations from the MSE comparison are as follows: (i) Compared with logspline, which uses the same link function but does not control the variation norm, HAL-MLE has substantially lower MSE on highly oscillatory DGPs while remaining competitive on the other DGPs. (ii) Compared with KDE (a linear smoother), HAL-MLE also attains much lower MSE on oscillatory, sudden-change, and heavy-tailed DGPs. (iii) TF(PP) and HAL-MLE are broadly comparable in MSE, because they are all MLEs on the same function class with different basis systems. TF(PP) loses some performance because our TF implementation in Section S.8 of the Supplementary Material (Hou et al., 2026) handles the normalizing constant in a relatively crude way. We acknowledge that the interpolation of the normalizing constant can be improved and is beyond the scope of this paper. Overall, the current TF/TFPP implementations with uniform knots appear more computationally efficient than HAL-MLE, because the divided-difference matrix is usually better conditioned than the truncated power basis matrix. These results support the plausibility of extending the theoretical properties from HAL-MLE to TF.

7. Case Study. This section presents a real data analysis using the classic Galaxies recession–velocity measurements. We follow the same dataset choice as Bak et al. (2021).

7.1. *Data and preprocessing.* We analyze the *Galaxies* data (R package `MASS`; Venables and Ripley, 2002), consisting of velocities for $n = 82$ galaxies in the Corona Borealis region measured from six conic sections of space (Postman, Huchra and Geller, 1986, Table 1). If galaxies are clustered, the velocity density is expected to exhibit three to seven modes that correspond to superclusters (Roeder, 1990). For the preprocessing, we apply a monotone affine rescaling of the observed velocities to $[0, 1]$. Throughout the figures, the horizontal axis represents the velocity as a proportion of the speed of light c: $x \in [0, 1]$ corresponds to v/c.

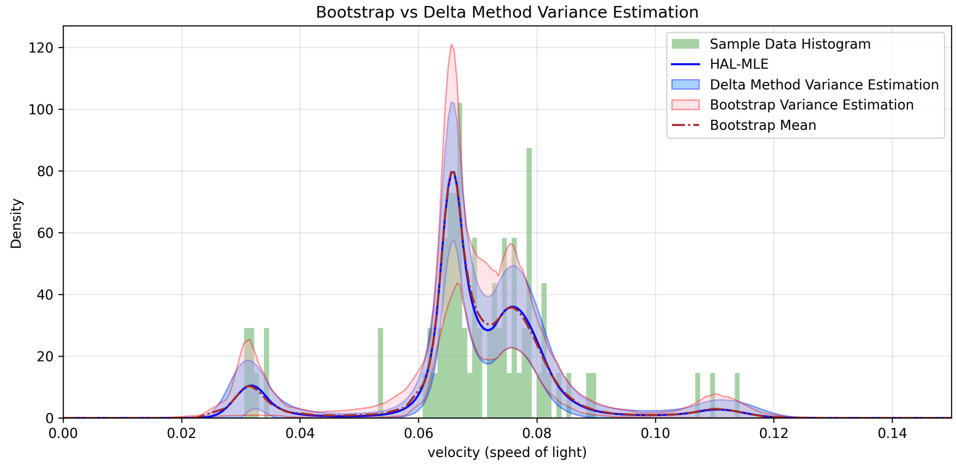

FIG 5. *Galaxy velocities: histogram of the data with 50 bins, HAL–MLE density, and variance estimation for the density via the delta method (blue band) compared with a nonparametric bootstrap (red band).*

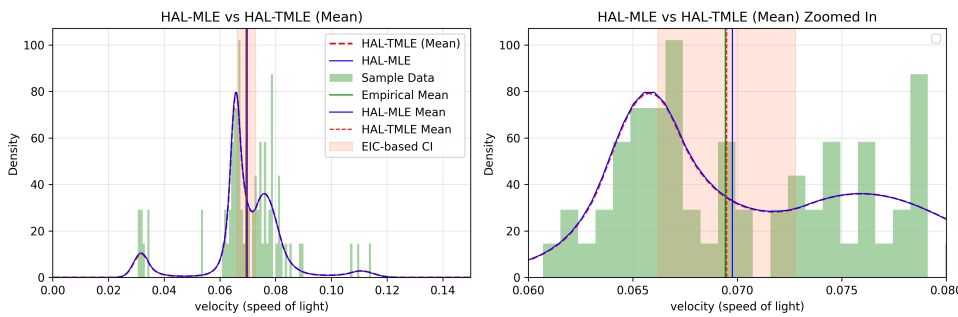

FIG 6. *Mean functional: plug–in HAL–MLE versus HAL–TMLE with the sample mean shown as the asymptotically efficient benchmark. Left: full scale; right: zoom around the dominant mode.*

7.2. *Density fit and variance estimation.* We fit the HAL–MLE density on the rescaled axis. The resulting fit captures all prominent modes and the troughs between them, reflecting the heterogeneous galaxy population. We estimate the standard errors for the density via the delta–method in Section 4.5. We benchmark the delta-method results against a nonparametric bootstrap. However, the bootstrap is much more computationally expensive, and thus we do not recommend it for practical use at this point. The shape of the density is similar to Bak et al. (2021) even though the data is rescaled.

Overall, the delta–method bands agree with the bootstrap, but there are regions where they are slightly narrower. One reason for this is that the parametric assumption of the Delta Method might cause under-estimation of variance. Another reason is that with the bootstrap, some bootstrap samples encounter the edge case of the solver, and the estimation is not accurate and varies a lot.

7.3. *Plug-in HAL-MLE versus HAL-TMLE.* We visualize the performance of the plug-in HAL-MLE and HAL-TMLE for the mean, median, and survival function. And we also visualize what targeting does to a fitted density based on the different statistical estimands. The results are shown below in Figure 6, Figure 7, and Figure 8.

The TMLE of the survival function at 0.5 shifts the entire tail after 0.5 up and down. It solves the EIC equation, but it breaks the smoothness of the density. The TMLE of mean and

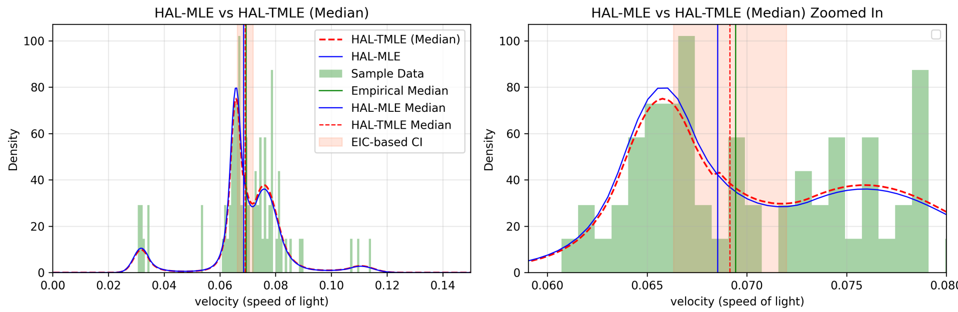

FIG 7. *Median functional: plug–in HAL–MLE versus HAL–TMLE with the sample median shown for reference. Targeting nudges the estimate to align with the efficient score for the quantile.*

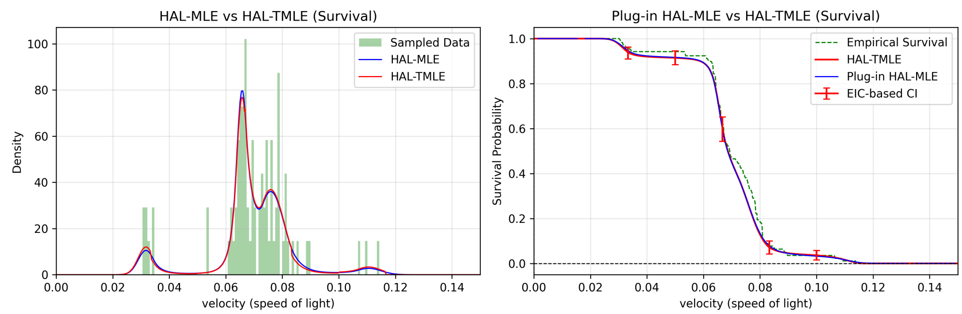

FIG 8. *Survival functional $S(t)$: plug–in HAL–MLE versus HAL–TMLE, compared to the empirical survival curve. Right panel overlays EIC–based intervals for representative t.*

median debiases the estimation from the plug-in HAL-MLE towards the sample mean and median, while still preserving a fitted density slightly different from the plug-in HAL-MLE.

8. Discussion. The limitation of this paper is that we only focused on the univariate case in order to compare and connect the classical density estimation methods in such setting. In the future, we can consider the generalization of the assumptions and extend the results to the multivariate case and other basis systems. We can also consider the adaptation to different data structures, such as censored/coarsened data.

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SUPPLEMENTARY MATERIAL

Supplement to "HAL-MLE Log-Splines Density Estimation (Part I: Univariate)" Contains all former appendix materials, including notation and technical definitions, proofs of the main theorems, additional simulation figures, DGP specifications, and optimization-algorithm details.

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